LUMS
A Not-for-Profit University

School of Science and Engineering

AI 501 Mathematics for Artificial Intelligence

ASSIGNMENT 1 – SOLUTIONS

Due Date: 1 pm, Sunday, October 12, 2025.
Format: 9 problems for a total of 100 marks
Instructions:

- You are allowed to collaborate with your peers but copying your colleague's solution is strictly prohibited. This is not a group assignment. Each student must submit his/her own assignment.
- Solve the assignment on blank A4 sheets and staple them before submitting.
- **There may be more than one correct solution for non-numeric questions.**
- Submit in-class or in the dropbox labeled AI-501 outside the instructor's office.
- **Write your name and roll no. on the first page.**
- Feel free to contact the instructor or the teaching assistants if you have any concerns.

- You represent the most competent individuals in the country, do not let plagiarism come in between your learning. In case any instance of plagiarism is detected, the disciplinary case will be dealt with according to the university's rules and regulations.
- We require you to acknowledge any use or contributions from generative AI tools. Include the following statement to acknowledge the use of AI where applicable.

  I have used [insert Tool Name] to [write, generate, plot or compute; explain specific use of generative AI] [number of times].

Problem 1 (10 marks)

Systems of Linear Equations and Null Spaces (Sensor Fusion in Robotics)

A mobile robot is trying to localize itself in a 2D environment. Its state is described by three variables:

$$x_1 = \text{horizontal position}, \quad x_2 = \text{vertical position}, \quad x_3 = \text{orientation}.$$

Three different sensor subsystems provide the following constraints:

- Sensor 1 reports that $x_1 - 2x_2 + 3x_3 = 1$.
- Sensor 2 reports that $-x_1 + 2x_2 - 4x_3 = -1$.
- Sensor 3 reports that $2x_1 - 4x_2 + 6x_3 = 2$.

(a) Construct the system of linear equations in matrix form $Ax = b$ from the above sensor constraints.

(b) Solve the system $\mathcal{L}(A, b)$. *Hint: Use systematic row operations to eliminate variables and reduce the system step by step.*

(c) Identify the free variables and determine the rank of A.

(d) Give the full set of solutions.

(e) Express the general solution as a sum of a particular solution w plus a vector u from Null(A).

Solution:

(a) Matrix form $Ax = b$. The coefficient matrix A, the unknown vector x, and the right-hand side b are

$$A = \begin{pmatrix} 1 & -2 & 3 \\ -1 & 2 & -4 \\ 2 & -4 & 6 \end{pmatrix}, \qquad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix},$$

so the system is $Ax = b$.

(b) Solve the system $\mathcal{L}(A, b)$ by row reduction. Write the augmented matrix $[A \mid b]$:

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 1 \\ -1 & 2 & -4 & -1 \\ 2 & -4 & 6 & 2 \end{array}\right].$$

Perform row operations:

1. $R_2 \leftarrow R_2 + R_1$:

$$R_2 : (-1+1,\ 2-2,\ -4+3,\ -1+1) = (0, 0, -1, 0).$$

2. $R_3 \leftarrow R_3 - 2R_1$:

$$R_3 : (2-2,\ -4+4,\ 6-6,\ 2-2) = (0, 0, 0, 0).$$

So the augmented matrix becomes

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Multiply the second row by -1 to get $x_3 = 0$:

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \Longrightarrow \quad x_3 = 0.$$

Substitute $x_3 = 0$ into the first row:

$$x_1 - 2x_2 + 3 \cdot 0 = 1 \quad \Longrightarrow \quad x_1 - 2x_2 = 1.$$

Conclusion of (b): The reduced system is

$$\begin{cases} x_1 - 2x_2 = 1, \\ x_3 = 0, \end{cases}$$

with one free variable x_2.

(c) Free variables and rank of A. There are $n = 3$ unknowns. The row-reduced coefficient matrix has two nonzero rows (the first and the second), so

$$\operatorname{rank}(A) = 2.$$

Thus the dimension of the null space is $n - \operatorname{rank}(A) = 3 - 2 = 1$. The free variable is x_2.

(d) Full set of solutions. Let $t \in \mathbb{R}$ be the free parameter $t = x_2$. Then

$$x_3 = 0, \qquad x_1 = 1 + 2x_2 = 1 + 2t.$$

Hence the full solution set is

$$\mathcal{L}(A, b) = \left\{ \begin{pmatrix} 1 + 2t \\ t \\ 0 \end{pmatrix} : t \in \mathbb{R} \right\}.$$

(e) Particular solution plus nullspace. Pick a particular solution by setting $t = 0$:

$$w = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

The homogeneous solutions (nullspace of A) correspond to $b = \mathbf{0}$. For the homogeneous system,

$$x_3 = 0, \qquad x_1 - 2x_2 = 0 \implies x_1 = 2x_2.$$

Let $s \in \mathbb{R}$ be the free parameter for the nullspace; then

$$u = s \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \qquad \text{so} \quad \operatorname{Null}(A) = \operatorname{span}\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

Therefore the general solution can be written as

$$x = w + u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \qquad s \in \mathbb{R},$$

which is equivalent to the parametric form found above with $s = t$.

Final answer: $\mathcal{L}(A, b) = \{(1 + 2t,\ t,\ 0)^T : t \in \mathbb{R}\}$, $\operatorname{rank}(A) = 2$, $\operatorname{Null}(A) = \operatorname{span}\{(2, 1, 0)^T\}$.

Problem 2 (10 marks)

Linear Independence, Singular Matrices, and Solutions of a SLE

In data science and AI, we often represent information as vectors in a feature space. For example, suppose you are building a model where the columns of a matrix A represent different feature directions. If these features are linearly independent, then they provide unique and non-redundant information. If they are dependent, some features can be expressed as combinations of others, leading to redundancy.

Similarly, when we try to fit data (represented by a vector b) into this feature space, the system of linear equations $\mathcal{L}(A, b)$ corresponds to finding a combination of features that explains the data. Whether this system is consistent depends on whether b lies in the span of the columns of A.

(a) Prove that if the column vectors of a square matrix A are linearly independent, then A is a non-singular (invertible) matrix.

(b) Prove that the system of linear equations $\mathcal{L}(A, b)$ is consistent $\iff$ b belongs to the span of the columns of A.

Solution: Throughout let A be an $n \times n$ matrix with columns $a_1, a_2, \ldots, a_n \in \mathbb{R}^n$.

(a) **Claim.** If the column vectors $a_1, \ldots, a_n$ of A are linearly independent, then A is non-singular (invertible).

Proof. The columns $a_1, \ldots, a_n$ are linearly independent in an n-dimensional vector space $\mathbb{R}^n$. Any set of n linearly independent vectors in $\mathbb{R}^n$ is a basis, so the columns span $\mathbb{R}^n$. Equivalently, the linear map

$$T : \mathbb{R}^n \to \mathbb{R}^n, \qquad T(x) = Ax$$

is surjective (its range is all of $\mathbb{R}^n$). For linear maps between finite-dimensional vector spaces of the same dimension, surjectivity implies injectivity. Thus T is injective, so the only solution of $Ax = 0$ is the trivial solution $x = 0$. Hence $\ker(A) = \{0\}$, so $\operatorname{rank}(A) = n$. A square matrix of full rank is invertible; equivalently $\det(A) \neq 0$. Therefore A is non-singular (invertible). ■

Alternate concise argument: linear independence of the n columns gives $\operatorname{rank}(A) = n$. For an $n \times n$ matrix rank n means invertible.

(b) **Claim.** The linear system $\mathcal{L}(A, b)$ (i.e. $Ax = b$) is consistent if and only if b belongs to the span of the columns of A.

Proof. Write the columns of A as $a_1, \ldots, a_n$. For any vector $x = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$ we have the identity (by matrix multiplication)

$$Ax = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n,$$

so Ax is precisely a linear combination of the columns of A.

$\Rightarrow$: If the system $Ax = b$ is consistent, there exists some x with $Ax = b$. By the displayed identity, b is a linear combination of the columns, so $b \in \operatorname{span}\{a_1, \ldots, a_n\}$.

$\Leftarrow$: Conversely, if $b \in \operatorname{span}\{a_1, \ldots, a_n\}$, then by definition there exist scalars $x_1, \ldots, x_n$ such that

$$b = x_1 a_1 + \cdots + x_n a_n,$$

and assembling these scalars into $x = (x_1, \ldots, x_n)^T$ gives $Ax = b$. Hence the system is consistent.

Therefore $\mathcal{L}(A, b)$ is consistent $\iff$ $b \in \operatorname{span}\{a_1, \ldots, a_n\}$. ■

Problem 3 (10 marks)

Vector Spaces, Subspaces, and Dimension

In some physical or sensing systems the response to a scalar stimulus differs by direction: one component responds linearly (keeping sign), while another responds only to the magnitude of the stimulus (losing sign information). Consider the set $V = \mathbb{R}^2$ with the usual vector addition $+$ but with a modified scalar multiplication

$$\alpha * \begin{pmatrix} x \\ y \end{pmatrix} := \begin{pmatrix} \alpha x \\ |\alpha|\, y \end{pmatrix} \qquad (\alpha \in \mathbb{R},\ (x, y) \in V).$$

This models an anisotropic scaling that discards the sign of the scalar on the second coordinate.

(a) Prove that $(V, +, *)$ satisfies all axioms of a vector space but one. State which one and provide an example.

Solution: We check the vector space axioms (over the field $\mathbb{R}$). Many of these are immediate because addition is the usual addition on $\mathbb{R}^2$.

(1) Additive closure, commutativity, associativity. These hold because addition is the usual addition in $\mathbb{R}^2$.

(2) Additive identity and inverses. The zero vector is $0_V = (0, 0)$ and for any $(x, y) \in V$ the additive inverse is $(-x, -y)$. These are the usual ones and satisfy the required properties.

(3) Compatibility of scalar multiplication with vector addition For $\alpha \in \mathbb{R}$ and $u = (x_1, y_1),\ v = (x_2, y_2) \in V$,

$$\alpha * (u + v) = \alpha * (x_1 + x_2,\ y_1 + y_2) = (\alpha(x_1 + x_2),\ |\alpha|(y_1 + y_2))$$

while

$$\alpha * u + \alpha * v = (\alpha x_1, |\alpha| y_1) + (\alpha x_2, |\alpha| y_2) = (\alpha(x_1 + x_2),\ |\alpha|(y_1 + y_2)).$$

Thus $\alpha * (u + v) = \alpha * u + \alpha * v$.

(4) Compatibility of scalar multiplication with field addition (this is the one that may fail). We compare $(\alpha + \beta) * v$ with $\alpha * v + \beta * v$. For $v = (x, y)$,

$$(\alpha + \beta) * v = ((\alpha + \beta)x,\ |\alpha + \beta| y),$$

whereas

$$\alpha * v + \beta * v = (\alpha x, |\alpha| y) + (\beta x, |\beta| y) = ((\alpha + \beta)x,\ (|\alpha| + |\beta|)y).$$

These are equal for all x, y exactly when $|\alpha + \beta| = |\alpha| + |\beta|$ for all $\alpha, \beta \in \mathbb{R}$, which is false in general (the triangle inequality is $|\alpha + \beta| \leq |\alpha| + |\beta|$, with strict inequality for many choices). Hence the distributive law over scalar addition generally fails.

A concrete counterexample: take $\alpha = 1,\ \beta = -1$ and $v = (0, 1)$. Then

$$(\alpha + \beta) * v = 0 * (0, 1) = (0, 0),$$

but

$$\alpha * v + \beta * v = 1 * (0, 1) + (-1) * (0, 1) = (0, 1) + (0, 1) = (0, 2).$$

So $(\alpha + \beta) * v \neq \alpha * v + \beta * v$. Therefore the axiom

$$(\alpha + \beta) * v = \alpha * v + \beta * v \quad \text{for all } \alpha, \beta \in \mathbb{R},\ v \in V$$

does not hold.

(5) Other scalar-multiplicative axioms that do hold.

- Associativity of scalars: $(\alpha\beta) * v = \alpha * (\beta * v)$. Indeed

  $$(\alpha\beta) * (x, y) = (\alpha\beta x,\ |\alpha\beta| y) = (\alpha\beta x,\ |\alpha||\beta| y)$$

  and

  $$\alpha * (\beta * (x, y)) = \alpha * (\beta x, |\beta| y) = (\alpha\beta x,\ |\alpha||\beta| y).$$

  So equality holds because $|\alpha\beta| = |\alpha||\beta|$.

- Identity scalar: $1 * v = (1 \cdot x, |1|y) = (x, y) = v$.
- 0-scalar property: $0 * v = (0, 0)$ is the zero vector.

Conclusion. The structure $(V, +, *)$ satisfies all vector space axioms except the distributivity of scalar multiplication over addition of scalars, i.e.

$$(\alpha + \beta) * v = \alpha * v + \beta * v$$

fails in general (counterexample shown above). Hence $(V, +, *)$ is not a vector space over $\mathbb{R}$.

In AI and robotics, polynomials are often used to approximate smooth trajectories or signals. For example, a cubic polynomial in $P_3(\mathbb{R})$ might describe the position of a robot arm over time, or approximate a nonlinear function in regression.

Now suppose we are interested in the set of cubic polynomials that satisfy two specific physical constraints:

$$W = \{p(x) \in P_3(\mathbb{R}) \mid p'(-1) = 0 \text{ and } p''(1) = 0\}.$$

Here:
$p'(-1) = 0$ means the trajectory has a horizontal slope at $x = -1$ (e.g., the robot is momentarily at rest).
$p''(1) = 0$ means the curvature at $x = 1$ is zero (e.g., the path is locally straight at that point).

(b) Prove or disprove that W is a linear subspace of $P_3(\mathbb{R})$.

(c) If it is a subspace, find a basis of W.

(d) If it is a subspace, determine the dimension of W.

Solution: Let $P_3(\mathbb{R})$ denote the vector space of real polynomials of degree at most 3. Write a general element of $P_3(\mathbb{R})$ as

$$p(x) = ax^3 + bx^2 + cx + d, \qquad a, b, c, d \in \mathbb{R}.$$

Then

$$p'(x) = 3ax^2 + 2bx + c, \qquad p''(x) = 6ax + 2b.$$

(b) Is W a linear subspace?

The set is

$$W = \{p \in P_3(\mathbb{R}) \mid p'(-1) = 0 \text{ and } p''(1) = 0\}.$$

The maps $p \mapsto p'(-1)$ and $p \mapsto p''(1)$ are linear functionals on $P_3(\mathbb{R})$ (they are evaluations of derivatives, which are linear operations). Thus

$$W = \ker\big(p \mapsto p'(-1)\big) \cap \ker\big(p \mapsto p''(1)\big),$$

an intersection of kernels of linear maps. Kernels of linear maps are subspaces, and an intersection of subspaces is a subspace. Hence W is a linear subspace of $P_3(\mathbb{R})$.

(Equivalently one can check closure under addition and scalar multiplication directly because the constraints are homogeneous linear equations.)

(c) Find a basis of W.

Impose the two constraints in terms of the coefficients a, b, c, d.

$$p'(-1) = 3a - 2b + c = 0, \qquad p''(1) = 6a + 2b = 0.$$

From $p''(1) = 0$ we get $6a + 2b = 0 \Rightarrow 3a + b = 0 \Rightarrow b = -3a$. Substitute into $p'(-1) = 0$:

$$3a - 2(-3a) + c = 3a + 6a + c = 9a + c = 0 \quad \Rightarrow \quad c = -9a.$$

The constant term d is unconstrained. Therefore every polynomial in W has the form

$$p(x) = ax^3 - 3ax^2 - 9ax + d = a\big(x^3 - 3x^2 - 9x\big) + d \cdot 1,$$

with parameters $a, d \in \mathbb{R}$.

Thus a convenient basis for W is

$$\boxed{\mathcal{B} = \{\, x^3 - 3x^2 - 9x,\ 1 \,\}},$$

since the two polynomials are linearly independent and span W.

(One may check quickly that both basis elements satisfy the constraints: for $q_1(x) = x^3 - 3x^2 - 9x$, $q_1'(-1) = 3(-1)^2 - 2 \cdot (-3) - 9 = 3 + 6 - 9 = 0$ and $q_1''(1) = 6 \cdot 1 + 2(-3) = 6 - 6 = 0$; for $q_2(x) = 1$, $q_2'(-1) = 0$, $q_2''(1) = 0$.)

(d) Dimension of W.

Since $\mathcal{B}$ contains two linearly independent vectors and spans W, we have

$$\dim W = 2.$$

Problem 4 (10 marks)

Matrix Inverse

Matrices represent transformations of data. For example: In computer graphics or robotics, a matrix A might transform coordinates from one frame to another. In machine learning, matrices are used to represent weight transformations or covariance structures.

Invertibility tells us whether a transformation can be reversed — i.e., whether we can recover the original data from the transformed data.

(a) Suppose $A^2 = 0$. Prove that A is not invertible. (Interpretation: this corresponds to a transformation that, when applied twice, sends every vector to zero — a process that cannot be reversed.)

(b) Prove that if A is invertible, then

$$(A^{-1})^T = (A^T)^{-1}.$$

(This identity is crucial in optimization, since transposed and inverse matrices frequently appear together in gradient updates and covariance analysis.)

Solution:

(a) Assume $A^2 = 0$. Suppose, for contradiction, that A is invertible. Then multiplying both sides of $A^2 = 0$ by A^{-1} yields

$$A = A^{-1}A^2 = A^{-1}0 = 0.$$

But then $A = 0$, which is not invertible. Contradiction. Hence A cannot be invertible.

(b) Start with the identity $AA^{-1} = I$. Taking transposes,

$$(AA^{-1})^T = I^T \quad \Rightarrow \quad (A^{-1})^T A^T = I.$$

Thus $(A^{-1})^T$ is a left inverse of A^T. Similarly,

$$(A^{-1}A)^T = I^T \quad \Rightarrow \quad (A^{-1})^T A^T = I.$$

So $(A^{-1})^T$ is also a right inverse. Therefore,

$$(A^{-1})^T = (A^T)^{-1}.$$

Problem 5 (10 marks)

Gram-Schmidt

We often want to simplify vector computations by working with orthogonal or orthonormal sets of vectors. For example, in Principal Component Analysis (PCA) and other dimensionality-reduction techniques, we want to re-express data in a basis where the directions are orthogonal. This makes projections easy to compute and interpretations cleaner.

However, most sets of vectors we encounter are not orthogonal to begin with. The Gram–Schmidt process helps us systematically transform a set of linearly independent vectors into an orthogonal (or orthonormal) set spanning the same subspace.

Consider the vectors

$$v_1 = \begin{pmatrix} -7 \\ -i \\ 0 \\ 1 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 2i \\ 3 \\ -1 \\ i \end{pmatrix} \in \mathbb{C}^4.$$

(a) In practice, data features are rarely orthogonal. Apply the Gram–Schmidt procedure to $\{v_1, v_2\}$ to produce an orthogonal set. (This demonstrates how we can start with arbitrary features and produce ones that are easier to use for projections and decompositions.)

(b) In high-dimensional spaces, we often want a full orthonormal basis to represent any possible vector. After orthogonalizing v_1 and v_2, determine whether it is possible to extend your orthonormal set to a full basis for $\mathbb{C}^4$. If so, construct such a basis. (This connects to building complete coordinate systems, e.g. for QR factorization in machine learning.)

(c) Subspaces defined by orthogonality conditions play a central role in machine learning, for example in defining null spaces and constraint spaces. Consider

$$W = \{v \in \mathbb{C}^4 \mid \langle v, v_1 \rangle = 0\}.$$

Determine whether W is a subspace of $\mathbb{C}^4$. (This shows how orthogonality naturally defines whole subspaces of valid solutions or constraints.)

Solution: We use the standard inner product on $\mathbb{C}^4$, $\langle u, v \rangle = \overline{u}^T v = \sum_{k=1}^4 \overline{u_k} v_k$.

$$v_1 = \begin{pmatrix} -7 \\ -i \\ 0 \\ 1 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 2i \\ 3 \\ -1 \\ i \end{pmatrix}.$$

(a) Gram–Schmidt to produce an orthogonal set.

Take $u_1 = v_1$. Compute the inner products

$$\langle v_1, v_1 \rangle = 51, \qquad \langle v_1, v_2 \rangle = -10i.$$

The Gram–Schmidt formula gives

$$u_2 = v_2 - \frac{\langle v_1, v_2 \rangle}{\langle v_1, v_1 \rangle} v_1 = v_2 + \frac{10i}{51} v_1.$$

Hence an orthogonal set is

$$u_1 = \begin{pmatrix} -7 \\ -i \\ 0 \\ 1 \end{pmatrix}, \qquad u_2 = \frac{1}{51} \begin{pmatrix} 32i \\ 163 \\ -51 \\ 61i \end{pmatrix},$$

and one may normalize to obtain an orthonormal pair

$$e_1 = \frac{u_1}{\sqrt{51}}, \qquad e_2 = \frac{u_2}{\|u_2\|},$$

where $\|u_2\| = \sqrt{\langle u_2, u_2 \rangle}$ (explicit but not simplified).

(b) Extend to a full orthonormal basis of $\mathbb{C}^4$.

The two orthonormal vectors e_1, e_2 span a 2-dimensional subspace; its orthogonal complement $(\text{span}\{e_1, e_2\})^\perp$ has dimension $4 - 2 = 2$, so we can find two vectors orthogonal to both e_1, e_2 and then orthonormalize them.

For example, solve $\langle w, e_1\rangle = \langle w, e_2\rangle = 0$. One convenient (un-normalized) solution basis is

$$w_1 = \begin{pmatrix} \frac{i}{23} \\ \frac{7}{23} \\ 1 \\ 0 \end{pmatrix}, \qquad w_2 = \begin{pmatrix} \frac{2}{23} \\ \frac{9i}{23} \\ 0 \\ 1 \end{pmatrix},$$

(which satisfy $\langle w_j, u_k\rangle = 0$ for $k = 1, 2$). Apply Gram–Schmidt to $\{w_1, w_2\}$ and normalize to obtain two orthonormal vectors e_3, e_4 with $\langle e_i, e_j\rangle = \delta_{ij}$. Then

$$\{e_1, e_2, e_3, e_4\}$$

is an orthonormal basis of $\mathbb{C}^4$ extending the orthonormalized $\{v_1, v_2\}$.

(Any other pair of independent vectors spanning $(\text{span}\{v_1, v_2\})^\perp$ could be used and orthonormalized; existence is guaranteed because $\dim((\text{span}\{v_1, v_2\})^\perp) = 2$.)

(c) Is $W = \{v \in \mathbb{C}^4 : \langle v, v_1\rangle = 0\}$ a subspace?

Yes. The map $f : \mathbb{C}^4 \to \mathbb{C},\ f(v) = \langle v, v_1\rangle$ is linear (over $\mathbb{C}$). Then

$$W = \ker f,$$

and the kernel of a linear map is always a subspace. Concretely, if $\langle v, v_1\rangle = \langle w, v_1\rangle = 0$ and $\alpha \in \mathbb{C}$, then $\langle v + w, v_1\rangle = 0$ and $\langle \alpha v, v_1\rangle = \alpha\langle v, v_1\rangle = 0$, so W is closed under addition and scalar multiplication.

□

Problem 6 (10 marks)

Chebyshev Inequality (Bounding Outliers in Data)

In machine learning and data analysis, we often work with high-dimensional vectors representing samples, features, or signals. A useful way to summarize the overall magnitude of a vector is its *root mean square (RMS) value*, which measures the average energy of the components.

The Chebyshev inequality gives us a way to bound the number of "large" entries in a vector: if the RMS is small, not many entries can be very large in magnitude. This is directly connected to detecting outliers in datasets and understanding variance in features.

(a) Suppose x is a 100-dimensional vector with $\text{rms}(x) = 1$. What is the maximum number of entries of x that can satisfy $|x_i| \geq 3$?

If your answer is k, explain why no such vector can have $k+1$ entries with absolute values at least 3. Then construct an explicit example of a 100-vector with RMS value 1 that achieves exactly k entries larger than 3 in absolute value.

(Interpretation: This part asks you to quantify how many extreme outliers are even possible, given the average energy of the vector.)

(b) Show that at least one entry of a vector must have absolute value at least as large as the RMS value of the vector.

(Interpretation: This guarantees the existence of at least one entry that is "representative" of the overall energy — no vector can have every entry below the RMS in magnitude.)

Solution:

(a) Let $x \in \mathbb{R}^{100}$ and $\text{rms}(x) = 1$. By definition

$$\text{rms}(x)^2 = \frac{1}{100}\sum_{i=1}^{100} x_i^2 = 1,$$

so

$$\sum_{i=1}^{100} x_i^2 = 100.$$

If k entries satisfy $|x_i| \geq 3$, then those k entries contribute at least $k \cdot 3^2 = 9k$ to the sum of squares. Hence

$$9k \leq \sum_{i=1}^{100} x_i^2 = 100,$$

so $k \leq \frac{100}{9}$. Since k is an integer, the maximum possible k is

$$k = \left\lfloor \frac{100}{9} \right\rfloor = 11.$$

To see why $k+1 = 12$ is impossible, note that if 12 entries had $|x_i| \geq 3$ then the sum of squares would be at least $12 \cdot 9 = 108 > 100$, contradicting $\sum_i x_i^2 = 100$.

We now construct an explicit 100-vector achieving $k = 11$ entries with $|x_i| > 3$. For example, take

$$x = (\underbrace{3.01, 3.01, \ldots, 3}_{11 \text{ entries}},\ 0.5822,\ 0, \ldots, 0),$$

i.e. 11 entries equal to 3.01, one entry equal to 0.582, and the remaining 88 entries equal to 0. The sum of squares is

$$11 \cdot 3.01^2 + 0.582^2 + 88 \cdot 0^2 = 11 \cdot 9 + 1 = 99 + 1 = 100,$$

so $\text{rms}(x) = \sqrt{100/100} = 1$, and exactly 11 entries satisfy $|x_i| > 3$.

(b) Let $x \in \mathbb{R}^n$ (here $n = 100$ but the argument works for any positive integer n) and write

$$\text{rms}(x) = \sqrt{\frac{1}{n}\sum_{i=1}^{n} x_i^2}.$$

Let $M = \max_{1 \leq i \leq n} |x_i|$. If, for contradiction, every entry had $|x_i| < \text{rms}(x)$, then $M < \text{rms}(x)$. But then

$$\sum_{i=1}^{n} x_i^2 \leq \sum_{i=1}^{n} M^2 \to nM^2 < n \cdot \text{rms}(x)^2 = \sum_{i=1}^{n} x_i^2,$$

a contradiction. Therefore $M \geq \mathrm{rms}(x)$, i.e. there exists at least one entry i with $|x_i| \geq \mathrm{rms}(x)$.

Problem 7 (10 marks)

Linear, Affine and Convex Combination We require you to plot the linear, affine, and convex combinations of vectors in $\mathbf{R}^3$. Write a Python script to generate three random vectors v_1, v_2, v_3, where each coordinate of these vectors is drawn from a standard normal distribution. We require you to submit both the code and the resulting plots.

Problem 8 (15 marks)

Regularized Demand Forecasting (Ridge Regression & Train/Test Evaluation)

An online retailer wants a stable linear model to forecast daily revenue from two marketing signals: number of site visitors (x_1) and paid-ad spend (x_2). Because these two signals are highly correlated in practice, ordinary least squares (OLS) can give large, unstable coefficients. Ridge regression (an L_2 penalty) is commonly used to stabilize the estimates and reduce variance at the cost of bias. Evaluating model performance on a held-out test set helps show whether regularization improves predictive performance in practice.

You are given the dataset below (each row is an observation):

obs	x_1 (visitors)	x_2 (ad spend)	y (revenue)
1	10	2	25
2	12	1	28
3	20	3	45
4	22	4	48
5	30	3	70
6	35	5	82

Let the linear model be

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2,$$

and define the design matrix $A \in \mathbb{R}^{6\times 3}$ containing a column of ones (intercept) and the two feature columns.

(a) **(Construct & OLS).** Construct the design matrix A and the response vector y. Write the OLS closed-form for $\widehat{\beta}_{\text{OLS}} = (A^\top A)^{-1} A^\top y$. (Show the shapes of matrices involved.) Compute $\widehat{\beta}_{\text{OLS}}$ numerically from the given data.
Why: this forces students to translate data into matrix form and perform the baseline estimation.

(b) **(Ridge solution & numerical comparison).** For ridge penalty parameter $\lambda = 10$, compute the ridge estimator

$$\widehat{\beta}_{\text{ridge}} = (A^\top A + \lambda I_3)^{-1} A^\top y.$$

(Note: do *not* regularize the intercept if you prefer — clearly state which convention you use.) Compute $\widehat{\beta}_{\text{ridge}}$ numerically.
Why: shows how regularization changes estimates when predictors are correlated.

(c) **(Train / Test MSE).** Use the first 4 observations (rows 1–4) as the training set and the final 2 observations (rows 5–6) as the test set. For both $\widehat{\beta}_{\text{OLS}}$ and $\widehat{\beta}_{\text{ridge}}$:

- compute the training mean squared error (MSE) and the test MSE,
- report which estimator gives lower test MSE and explain why (in terms of bias/variance and the effect of collinearity).

Why: demonstrates the practical value of regularization for generalization.

(d) **(Conditioning and interpretation).** Compute the condition number of $A^\top A$ (ratio of largest to smallest eigenvalue). Explain briefly (2–3 sentences) how a large condition number justifies using ridge regression in this problem.
Why: connects algebraic properties (conditioning) to numerical instability and regularization.

Solution:

(a) Construct A and OLS

The design matrix $A \in \mathbb{R}^{6\times 3}$ (column of ones for intercept plus two features) and response vector $y \in \mathbb{R}^6$ are

$$A = \begin{bmatrix} 1 & 10 & 2 \\ 1 & 12 & 1 \\ 1 & 20 & 3 \\ 1 & 22 & 4 \\ 1 & 30 & 3 \\ 1 & 35 & 5 \end{bmatrix}, \qquad y = \begin{bmatrix} 25 \\ 28 \\ 45 \\ 48 \\ 70 \\ 82 \end{bmatrix}.$$

Shapes: A is 6×3, $A^\top A$ is 3×3, $A^\top y$ is 3×1, and the OLS solution is

$$\widehat{\beta}_{\text{OLS}} = (A^\top A)^{-1} A^\top y, \qquad \widehat{\beta}_{\text{OLS}} \in \mathbb{R}^3.$$

Compute the necessary matrices (numerically):

$$A^\top A = \begin{bmatrix} 6 & 129 & 18 \\ 129 & 3253 & 445 \\ 18 & 445 & 64 \end{bmatrix}, \qquad A^\top y = \begin{bmatrix} 298 \\ 7512 \\ 1025 \end{bmatrix}.$$

$$(A^\top A)^{-1} = \begin{bmatrix} 1.18 & -0.02865 & -0.1338 \\ -0.02865 & 0.006988 & -0.04053 \\ -0.1338 & -0.0405 & 0.335 \end{bmatrix}, \qquad A^\top y = \begin{bmatrix} 298 \\ 7512 \\ 1025 \end{bmatrix}.$$

Hence

$$\widehat{\beta}_{\text{OLS}} = \begin{bmatrix} 0.47624039 \\ 2.41229909 \\ -0.89133473 \end{bmatrix} \approx \begin{bmatrix} 0.4763 \\ 2.4123 \\ -0.8913 \end{bmatrix}.$$

(So the fitted model is $\hat{y} = 0.4763 + 2.4123\, x_1 - 0.8913\, x_2$.)

(b) Ridge ($\lambda = 10$); convention

We use ridge with penalty parameter $\lambda = 10$. **Convention used:** do *not* regularize the intercept (i.e., only penalize the two slope coefficients). This corresponds to

$$\widehat{\beta}_{\text{ridge}} = \left(A^\top A + \lambda D\right)^{-1} A^\top y,$$

where $D = \text{diag}(0, 1, 1)$.

$$(A^\top A + \lambda D)^{-1} = \begin{bmatrix} 1.127 & -0.0398 & -0.0345 \\ -0.0398 & 0.0031 & -0.009 \\ -0.0345 & -0.009 & 0.076 \end{bmatrix}, \qquad A^\top y = \begin{bmatrix} 298 \\ 7512 \\ 1025 \end{bmatrix}.$$

Numerically (with $\lambda = 10$):

$$\widehat{\beta}_{\text{ridge}} = \begin{bmatrix} 1.13001867 \\ 2.25676937 \\ 0.00536881 \end{bmatrix} \approx \begin{bmatrix} 1.1300 \\ 2.2568 \\ 0.0054 \end{bmatrix}.$$

(With ridge the coefficient on x_2 is shrunk dramatically toward zero compared with OLS; the intercept changed because we did not penalize it.)

Alternatively, regularising the bias as well:

$$\widehat{\beta}_{\text{ridge}} = \left(A^\top A + \lambda I\right)^{-1} A^\top y,$$

where $D = \text{diag}(0, 1, 1)$.

$$(A^\top A + \lambda I)^{-1} = \begin{bmatrix} 0.0918 & -0.0032 & -0.0028 \\ 0.0032 & 0.0018 & -0.0101 \\ -0.0028 & -0.0101 & 0.075 \end{bmatrix}, \qquad A^\top y = \begin{bmatrix} 298 \\ 7512 \\ 1025 \end{bmatrix}.$$

Numerically (with $\lambda = 10$):

$$\widehat{\beta}_{\text{ridge}} = \begin{bmatrix} 0.092 \\ 2.29 \\ 0.037 \end{bmatrix}$$

(c) Train / Test MSE

Use rows 1–4 as training set and rows 5–6 as test set. Let $A_{\text{train}}, y_{\text{train}}$ be the first 4 rows and $A_{\text{test}}, y_{\text{test}}$ the last 2 rows. For any estimator β we compute

$$\text{MSE} = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2,$$

with n equal to 4 for training and 2 for test here.

Numerical MSEs (rounded to 4 significant figures):

- $\widehat{\beta}_{\text{OLS}}$:

$$\text{Train MSE}_{\text{OLS}} \approx 2.519, \qquad \text{Test MSE}_{\text{OLS}} \approx 1.216.$$

- $\widehat{\beta}_{\text{ridge}}$ ($\lambda = 10$, intercept unpenalized):

$$\text{Train MSE}_{\text{ridge}} \approx 2.800, \qquad \text{Test MSE}_{\text{ridge}} \approx 2.385.$$

Which estimator gives lower test MSE? In this dataset $\widehat{\beta}_{\text{OLS}}$ gives the lower test MSE (about 1.216 vs. 2.385 for ridge with $\lambda = 10$). Thus, for this particular split and this λ, OLS generalizes slightly better.

Explanation (bias/variance and collinearity): Ridge regression trades variance for bias: by shrinking coefficients (especially along directions corresponding to small eigenvalues of $A^\top A$) it reduces estimator variance at the cost of bias. If multicollinearity causes very large-variance OLS coefficients, ridge often improves test error. However, whether ridge helps depends on the data split and the chosen λ: here the chosen $\lambda = 10$ produced enough bias (not enough variance reduction relative to OLS) that test MSE increased. With a different λ (smaller) or different train/test split, ridge might lower test MSE. The result is consistent with the general principle: when collinearity-induced variance dominates, ridge helps; otherwise it can hurt due to added bias.

(d) Conditioning of $A^\top A$

Compute the eigenvalues of $A^\top A$. Numerically they are approximately

$$\lambda_1 \approx 3319.05, \quad \lambda_2 \approx 3.1176, \quad \lambda_3 \approx 0.82978.$$

The condition number (ratio largest/smallest eigenvalue) is

$$\kappa(A^\top A) = \frac{\lambda_{\max}}{\lambda_{\min}} \approx \frac{3319.05}{0.82978} \approx 3999.9 \approx 4.0 \times 10^3.$$

Interpretation (2–3 sentences): A large condition number (here ≈ 4000) means $A^\top A$ is ill-conditioned: small changes in the data or small numerical errors can produce large changes in the OLS solution. This instability is typically caused by strong collinearity between predictors (here x_1 and x_2 are highly correlated). Ridge regression improves conditioning by adding λI (or λD), increasing the smallest eigenvalues and thereby stabilizing the inverse at the cost of introducing bias.

Problem 9 (15 marks)

A hospital deploys a logistic regression model to estimate the probability that incoming patients have a serious condition ("positive"). Clinical decisions have asymmetric costs: missing a true positive (false negative) is typically much more costly than a false alarm (false positive). Thus, choosing a decision threshold matters; so does understanding precision/recall tradeoffs and the ROC/AUC.

You are given predicted probabilities from a trained logistic model and the true labels for 10 patients:

patient	1	2	3	4	5	6	7	8	9	10
$\hat{p}$	0.95	0.89	0.76	0.65	0.55	0.45	0.35	0.25	0.15	0.05
y	1	1	1	0	1	0	0	1	0	0

(There are five true positives and five true negatives.)

(a) **(Threshold 0.5 — metrics).** Using threshold $t = 0.5$ (predict positive when $\hat{p} \geq 0.5$), compute:

- the confusion matrix (TP, FP, FN, TN),
- accuracy, precision, recall (sensitivity), specificity, and F1 score.

Why: shows how thresholding maps probabilities to discrete decisions and how different metrics measure different errors.

(b) **(Cost-sensitive decision).** Suppose the clinical cost of a false negative is $C_{\text{FN}} = 10$ and the cost of a false positive is $C_{\text{FP}} = 1$ (correct predictions cost 0).

(i) Derive the optimal decision rule (threshold in terms of C_{FP} and C_{FN}) that minimizes expected cost for a single patient given predicted probability $\hat{p}$.

(ii) Using the derived threshold, classify the 10 patients and compute the resulting confusion matrix and total expected cost (sum of realized misclassification costs).

Why: connects probabilistic outputs to operational decisions under asymmetric costs.

(c) **(ROC & AUC).** Construct the ROC curve points by sweeping the threshold through the distinct predicted probabilities (include thresholds just above each distinct $\hat{p}$ and just below), compute the true positive rate (TPR) and false positive rate (FPR) at each step, and approximate the Area Under the Curve (AUC) using the trapezoidal rule. Report the AUC numeric value.
Why: AUC summarizes rank quality of the probabilistic model across thresholds.

Solution:

(a) Threshold $t = 0.5$ — confusion matrix and metrics

With threshold $t = 0.5$ we predict $\hat{y}_i = \mathbf{1}\{\hat{p}_i \geq 0.5\}$. The predicted positives are patients $1, 2, 3, 4, 5$ (their $\hat{p} \geq 0.5$); predicted negatives are $6, 7, 8, 9, 10$.

Compare to true labels:

- Predicted positives $\{1, 2, 3, 4, 5\}$ have true labels $\{1, 1, 1, 0, 1\}$ $\Rightarrow$ 4 true positives (TP) and 1 false positive (FP).
- Predicted negatives $\{6, 7, 8, 9, 10\}$ have true labels $\{0, 0, 1, 0, 0\}$ $\Rightarrow$ 1 false negative (FN) and 4 true negatives (TN).

Thus the confusion matrix entries are

$$\boxed{\text{TP} = 4,\ \text{FP} = 1,\ \text{FN} = 1,\ \text{TN} = 4}$$

Now compute common metrics (using denominator conventions that avoid division by zero):

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{N} = \frac{4+4}{10} = 0.8,$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{4}{4+1} = 0.8,$$

$$\text{Recall (Sensitivity)} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{4}{4+1} = 0.8,$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} = \frac{4}{4+1} = 0.8,$$

$$\text{F1 score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot 0.8 \cdot 0.8}{0.8 + 0.8} = 0.8.$$

All five of these metrics evaluate to 0.8 for this threshold.

(b) Cost-sensitive decision

(i) Optimal decision rule (threshold) in terms of costs

For a single patient with predicted probability $\hat{p} = P(y = 1 \mid x)$, the expected cost of predicting *positive* is the cost incurred when the patient is actually negative:

$$\mathbb{E}[\text{cost} \mid \text{predict positive}] = C_{\text{FP}} \cdot P(y = 0 \mid x) = C_{\text{FP}}(1 - \hat{p}).$$

The expected cost of predicting *negative* is the cost when the patient is actually positive:

$$\mathbb{E}[\text{cost} \mid \text{predict negative}] = C_{\text{FN}} \cdot P(y = 1 \mid x) = C_{\text{FN}}\,\hat{p}.$$

Predict positive whenever its expected cost is no larger than predicting negative:

$$C_{\text{FP}}(1 - \hat{p}) \;\le\; C_{\text{FN}}\hat{p} \quad\Longrightarrow\quad C_{\text{FP}} \le \hat{p}\,(C_{\text{FN}} + C_{\text{FP}}) \quad\Longrightarrow\quad \boxed{\hat{p} \ge \frac{C_{\text{FP}}}{C_{\text{FP}} + C_{\text{FN}}}}.$$

Thus the optimal threshold is

$$t^\star \;=\; \frac{C_{\text{FP}}}{C_{\text{FP}} + C_{\text{FN}}}.$$

(ii) Apply with $C_{\text{FN}} = 10$, $C_{\text{FP}} = 1$

Plugging the numbers:

$$t^\star = \frac{1}{1+10} = \frac{1}{11} \approx 0.090909.$$

So we predict positive whenever $\hat{p} \ge 0.090909$. For our 10 patients this means everyone with $\hat{p} \ge 0.15$ (patients 1–9) is predicted positive; only patient 10 ($\hat{p} = 0.05$) is predicted negative.

Compare to truth:

- Predicted positives: patients $1, 2, 3, 4, 5, 6, 7, 8, 9$ with true labels $\{1, 1, 1, 0, 1, 0, 0, 1, 0\} \Rightarrow \text{TP} = 5$ (patients 1,2,3,5,8) and $\text{FP} = 4$ (patients 4,6,7,9).
- Predicted negatives: patient 10 with true label $0 \Rightarrow \text{TN} = 1$ and $\text{FN} = 0$.

Confusion matrix under cost-optimal rule:

$$\boxed{\text{TP} = 5,\ \text{FP} = 4,\ \text{FN} = 0,\ \text{TN} = 1}$$

Total realized cost (sum of misclassification costs):

$$\text{Total cost} \;=\; C_{\text{FP}} \cdot \text{FP} + C_{\text{FN}} \cdot \text{FN} \;=\; 1 \cdot 4 + 10 \cdot 0 \;=\; \boxed{4}.$$

This decision rule eliminated all false negatives (which were costly) at the expense of several false positives; the total cost is 4 under the chosen asymmetry.

(c) ROC curve points and AUC (trapezoidal rule)

We construct the ROC by varying threshold t and computing

$$\mathrm{TPR}(t) = \frac{\mathrm{TP}(t)}{\#\text{positives}}, \qquad \mathrm{FPR}(t) = \frac{\mathrm{FP}(t)}{\#\text{negatives}}.$$

There are 5 positives and 5 negatives. Sort predictions descending (they are already sorted):

patient	1	2	3	4	5	6	7	8	9	10
$\hat{p}$	0.95	0.89	0.76	0.65	0.55	0.45	0.35	0.25	0.15	0.05
y	1	1	1	0	1	0	0	1	0	0

We sweep thresholds from above the largest score down to below the smallest. The ROC vertices (in order of decreasing threshold) are:

step	$(\mathrm{FPR}, \mathrm{TPR})$	reason
0	(0.0, 0.0)	threshold above 0.95 (no positives)
1	(0.0, 0.2)	only patient 1 positive (TP=1,FP=0)
2	(0.0, 0.4)	patients 1–2 positive (TP=2,FP=0)
3	(0.0, 0.6)	patients 1–3 positive (TP=3,FP=0)
4	(0.2, 0.6)	add patient 4 (negative) (FP=1)
5	(0.2, 0.8)	add patient 5 (positive) (TP=4)
6	(0.4, 0.8)	add patient 6 (negative) (FP=2)
7	(0.6, 0.8)	add patient 7 (negative) (FP=3)
8	(0.6, 1.0)	add patient 8 (positive) (TP=5)
9	(0.8, 1.0)	add patient 9 (negative) (FP=4)
10	(1.0, 1.0)	all predicted positive (FP=5)

Plotting FPR on the x-axis and TPR on the y-axis, we obtain the piecewise linear ROC joining the points above.

AUC by trapezoidal rule. Compute the area under the ROC by summing trapezoids between consecutive $(\mathrm{FPR}, \mathrm{TPR})$ points. The nonzero trapezoid contributions occur when FPR increases:

$$\begin{aligned}
\text{area} &= (0.2 - 0.0) \cdot \frac{0.6 + 0.6}{2} \;+\; (0.4 - 0.2) \cdot \frac{0.8 + 0.8}{2} \\
&\quad + (0.6 - 0.4) \cdot \frac{0.8 + 0.8}{2} \;+\; (0.8 - 0.6) \cdot \frac{1.0 + 1.0}{2} \\
&\quad + (1.0 - 0.8) \cdot \frac{1.0 + 1.0}{2} \\
&= 0.2 \cdot 0.6 + 0.2 \cdot 0.8 + 0.2 \cdot 0.8 + 0.2 \cdot 1.0 + 0.2 \cdot 1.0 \\
&= 0.12 + 0.16 + 0.16 + 0.20 + 0.20 = \boxed{0.84}.
\end{aligned}$$

(As a check, AUC can be computed from pairwise ranking: it equals the fraction of positive–negative pairs where the positive has a higher score than the negative. Counting pairs here gives 21 favorable out of 25 total pairs, so AUC $= 21/25 = 0.84$, in agreement.)

— End of Assignment —

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	15	
9	15	
Total:	100	